

Project : Security Alert Monitoring & Incident Response Automation

Project Overview:

Built an automated security alert monitoring and incident response workflow using n8n to receive security events through webhooks, classify security risks, collect IP-related information, maintain structured incident logs, and send automated email alerts for high-risk security activities.

Tools & Technologies:

- n8n Workflow Automation
- Webhook
- IF Condition Logic
- Edit Fields
- HTTP Request / API
- Google Sheets Integration
- Gmail Integration
- JSON Data Handling

Workflow Explanation:

- Webhook receives security event data from an external source.
- IF Condition evaluates the security event and determines the appropriate workflow path based on risk level.
- Edit Fields prepares and structures the security alert information.
- HTTP Request retrieves IP-related information such as location and country details.
- Google Sheets stores the security event as a structured incident record.
- Alert information includes Time, Event, User, IP Address, Risk, Location, Status, Action Taken, Priority, Alert ID, Assigned To, Investigation Notes, and Resolution.
- Gmail sends an automated high-risk security alert notification to the designated security email.

Workflow Structure:

Webhook



IF Condition (Risk Classification)



Edit Fields (Alert Data Preparation)



HTTP Request (IP Information Lookup)



Google Sheets (Incident Log)



Gmail (Security Alert Notification)

Features:

- Automated security event monitoring
- Webhook-based security event collection
- Risk-based workflow classification
- IP information lookup
- Automated incident logging
- Unique security Alert ID generation
- Incident status tracking
- Priority classification
- Investigation notes management
- Assigned security team tracking
- Automated high-risk email alerts
- Centralized security incident records
- Business process automation using n8n

Testing Results:

High-Risk Security Alert Test:

- A high-risk security event was submitted through the webhook.
- The workflow successfully processed the event.
- IP information was retrieved successfully.
- The incident was recorded in Google Sheets.
- A high-risk security alert email was successfully delivered through Gmail.

Low-Risk Security Alert Test:

- A low-risk security event was submitted through the webhook.
- The workflow successfully processed the low-risk event.
- The event was recorded in the appropriate Google Sheets path.

- The low-risk workflow behavior was verified successfully.

Skills Demonstrated:

- Workflow Automation
- Webhook Integration
- Conditional Logic
- HTTP/API Integration
- JSON Data Handling
- Security Event Monitoring
- Risk Classification
- Incident Logging
- Email Automation
- Google Sheets Integration
- n8n Workflow Development
- Basic Incident Response Concepts

Stack:

- n8n
- Webhooks
- HTTP Request/API
- Google Sheets
- Gmail API/Integration
- JSON
- Automation Workflows